1.3 Rivers present opportunities and hazards for people

This chapter will explain:

★ the opportunities of living near a river
★ the hazards of living near a river: floods and pollution
★ detailed specific example: the 2019 Mississippi River flood
★ detailed specific example: restoring Shanghai's Suzhou Creek.

The opportunities of living near a river

The land surrounding rivers, particularly the lower courses of rivers, is often very densely populated. This is because of:

» the significant supply of water that can be obtained from a large river
» the fertile soils usually found on floodplains, which allow for highly productive agriculture
» the opportunities for fishing, which provides a substantial additional source of food
» the use of the river as an important transport route
» the large areas of flat land for construction of homes and infrastructure.

The emergence of the world's first cities some 5500 years ago is the result of the advantages listed above. The areas that first experienced this important socioeconomic change towards larger urban areas were:

» Mesopotamia – the valleys of the Tigris and Euphrates Rivers (modern-day Iraq)
» the lower Nile valley (Egypt, Figure 1.34)
» the floodplains of the River Indus (Pakistan).

▲ **Figure 1.34** Arable farming in the lower Nile valley, with the pyramids in the background

Rivers provide sources of power, mainly in terms of producing hydroelectricity at a range of scales, along with opportunities for the development of tourism, which can be a major source of employment. Many major river infrastructure projects have multiple purposes (Figure 1.35).

Locations along or near rivers frequently provide attractive environments in which to live. This fact is often reflected in higher house prices close to rivers compared with further away. However, in many areas, particularly those that have been subject to recent floods, houses have proved difficult to sell and also to insure.

▲ **Figure 1.35** A system of locks along the River Danube – part of a multipurpose river scheme providing hydroelectricity, flood control and improved navigation

The hazards of living near a river

Floods

Floods are a natural feature of all rivers. For most of the time a river is contained within its channel, but at other times it may burst its bank and a flood occurs. Floods bring advantages such as water and fertile alluvium, which allow farmers to grow crops. But the problem is that they may bring too much water and too much silt. The results can be devastating, as the many disastrous floods throughout the history of China show. Rapid river bank erosion during floods can cause population displacement and socioeconomic impacts. For example, the flooding of the Meghna River in Bangladesh caused major disruptions during the 1990s and 2000s. The frequency and intensity of floods has increased significantly in many countries due to climate change, and this is a trend that is likely to continue.

Hazards associated with flooding can be divided into primary, secondary and tertiary effects.

Primary effects:

» Lives being lost, particularly when flooding is severe and occurs with little warning.
» Rural areas being inundated, resulting in crop loss and livestock destruction.
» Urban areas being inundated, impacting housing, industry and services infrastructure. Underground transit systems can be particularly affected.
» Flood control and other river infrastructure being damaged – bridges, levées, flood walls.
» Pollutants being transported and distributed over a wide area.

Secondary effects:

» Drinking water supply systems becoming polluted, especially if sewage treatment plants are flooded.
» Disease spreading due to contaminated water and other factors.
» Electricity and other energy-supply services being disrupted.
» Transportation systems being impacted, which may result in food shortages.

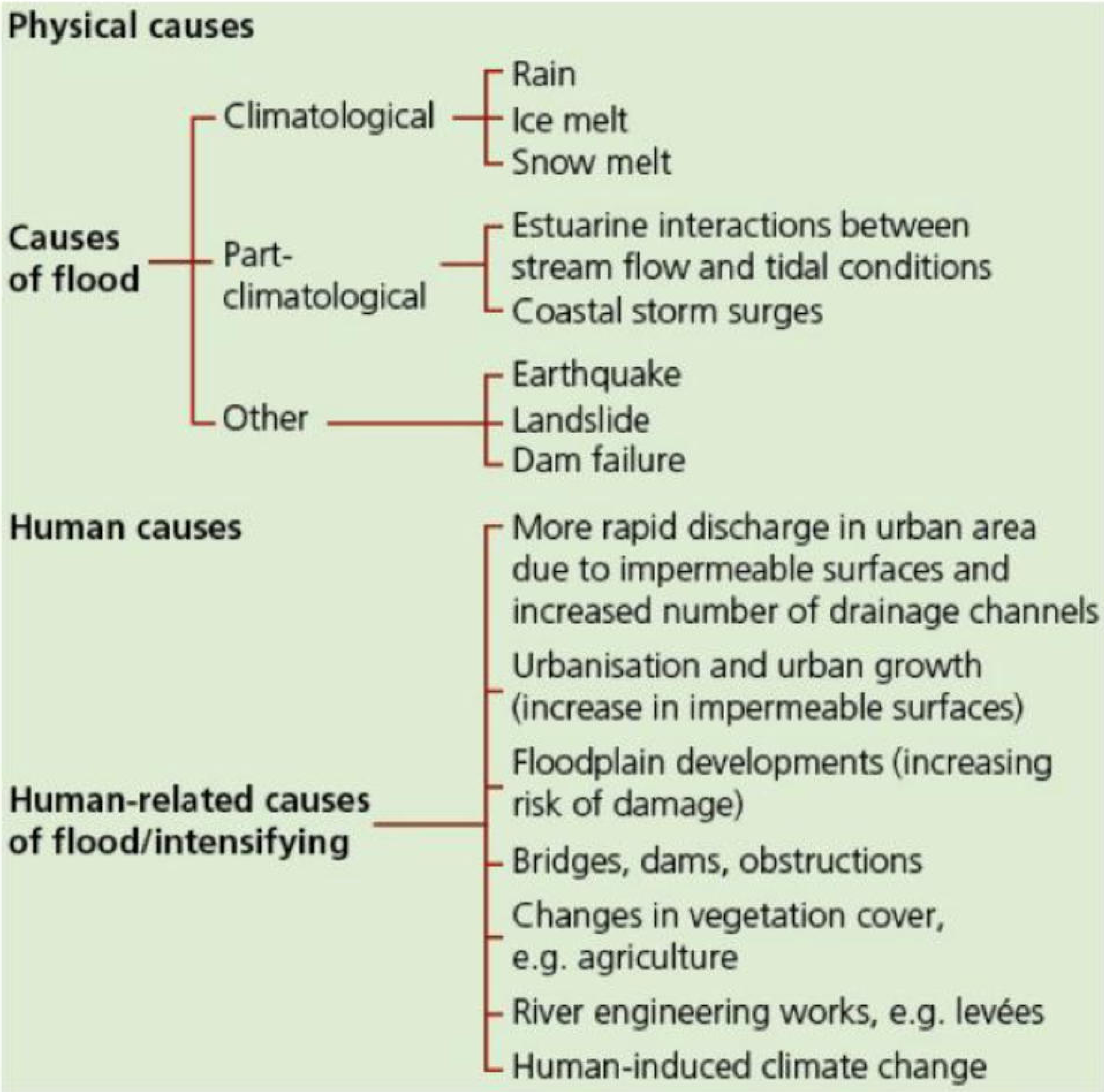
Tertiary (long-term) effects:

» Wildlife habitats being destroyed.
» The routes of river channels may change, leaving old channels dry and infrastructure abandoned.
» The quality of some farmland may be adversely impacted, leading to farming being abandoned or severely curtailed.
» Land, housing and all types of infrastructure may become more expensive to insure.

Floods account for about one-third of all natural catastrophes. They cause more than half the fatalities and are responsible for one-third of the economic losses.

The natural causes of flooding

The initial causes of floods are natural. However, human interference intensifies many floods (Figure 1.36). A flood is a high flow of water that overtops the banks of a river. The primary causes of floods are mainly the results of external climatic forces. The secondary causes of floods tend to be drainage-basin specific. Most floods in Western Europe, for example, are associated with deep depressions (low-pressure systems) in autumn and winter, which are both long in duration and wide in areal coverage. By contrast, in India up to 70 per cent of the annual rainfall occurs in just 100 days, in the summer southwest **monsoon**. Elsewhere, melting snow may be responsible for widespread flooding.



▲ **Figure 1.36** The natural and human causes of floods

Heavy rainfall is the major cause of flooding in most countries, particularly long periods of rain that completely saturate the soil. When soil is saturated, surface runoff (overland flow) is at its most significant, resulting in the extremely rapid movement of water into river systems. Snowmelt can be an important contributing factor in some areas, as unusually rapid snowmelt can sometimes be the major cause of floods.

Floods caused primarily by heavy rainfall are sometimes referred to as slow-building floods, as opposed to **flash floods** that impact much more suddenly. Flash floods are usually the result of a short period of unusually intensive rainfall. They are especially common in arid and semi-arid areas, and sometimes occur after a period of **drought** when land has been baked hard by the sun, preventing infiltration. Small river basins are particularly subject to flash floods.

In low-lying areas of active floodplains and river estuaries, flood water can spread out easily over a large area, but the gradient is too small to allow the water to leave the area quickly. In Bangladesh, 110 million people are living relatively unprotected on the floodplains of the Ganges, Brahmaputra and Meghna Rivers. Floods caused by the monsoon regularly cover 20–30 per cent of the flat delta. In very serious floods, up to half of the country may be flooded.

Flood-intensifying conditions include a range of factors that alter the response of a drainage basin to a given storm. These factors include the topography, vegetation, soil type, rock type and specific characteristics of the drainage basin. The potential for damage by flood waters increases substantially with velocity and speeds above 3 m per second, and can undermine the foundations of buildings. The physical stresses on buildings are increased even more, probably by hundreds of times, when the rough, rapidly flowing water contains debris such as rocks, sediment and trees.

The human causes of river flooding

Urbanisation and urban growth

The creation of highly impermeable surfaces reduces the volume of water that can infiltrate into the ground and therefore increases runoff. A dense network of drains and sewers increases drainage density so runoff is transferred quickly into river channels. Increased storm runoff means many sewerage systems cannot cope with the resulting peak flow. Natural river channels are often constricted by bridge supports or riverside facilities, reducing their carrying capacity.

Floodplain development

The increasing development of floodplains over the past century for housing and economic activity at the expense of farming and forestry, has significantly increased the proportion of land covered by impermeable surfaces, putting a higher percentage of people at risk of flooding in many countries.

The build-up of river debris from human activities

Debris from industrial, agricultural and domestic activities can significantly decrease the flow rates of river channels. Such debris can also pollute water channels and impact ecosystems.

Deforestation and poor agricultural practices

Deforestation, particularly in upland areas, has increased the rapid movement of water into lowland areas, often overwhelming the capacity of the physical landscape to absorb such large amounts of water. Poor agricultural practices, such as removing hedgerows and leaving large areas bare of vegetation, increase the problem of the rapid movement of water into river channels.

Human-induced climate change

A warmer atmosphere can hold more moisture and subsequently release greater volumes of precipitation. Higher temperatures can cause more ‘rain-on-snow’ events, with warm rains causing faster and often earlier snowmelt. Hurricanes and other storm events are predicted to become more frequent.

Activities

1 Discuss two physical causes of flooding.
2 How has urban and economic development on floodplains increased the risk of flooding?
3 Outline the causes and consequences of a major flood that has occurred anywhere in the world in the last few years.

The impacts of river flooding

Floods cause more than US$40 billion in damage worldwide every year. Floods are projected to increase for many world regions. The USA, Australia, South Africa and India were among countries affected by serious flooding in 2022 (Table 1.3).

▼ **Table 1.3** Examples of high-impact flood events in 2022

Event	Effects
Pakistan floods	● Extremely heavy monsoon rains impacted more than 33 million people. ● The month of July experienced 181 per cent of average precipitation and August 243 per cent. ● The number of deaths exceeded 1700 and there were economic losses of nearly US$30 billion.
India floods	● Monsoon flooding and landslides caused 700 deaths.
Bangladesh floods	● Floods affected 7 million people and 141 people died.
Sahel floods	● Nigeria, Niger, Chad and southern Suden all suffered floods due to heavy rain at the end of the monsoon season (Oct-Dec). ● Deaths exceeded 600 in Nigeria and there were economic losses of US$4.2 billion.

Climate change is altering the frequency, severity and location of flooding. A higher average global temperature is increasing the amount and intensity of rainfall during precipitation events. This is likely to amplify the severity of flooding, although floods could become rarer in some regions.

Flooding can seriously affect the quality of groundwater and surface water resources, along with the quantity of safe water that can be delivered to water users. Damaged infrastructure can result in severe interruptions to water supply services. The most frequent problems are:

» scarcity of safe drinking water
» disruption of **water treatment** facilities
» the outbreak of disease as a consequence.

The reduction of groundwater quality can be caused by pollutants being transported below the surface on groundwater recharge. Flooding can affect **wellfields**, with floodwater contaminating damaged wells. A wellfield is the land above and surrounding the wells that have been drilled into an aquifer. Significant interruptions to treatment facilities may occur, along with damage to distribution systems. Severe flooding can result in the interruption of abstraction from artificial reservoirs and the deterioration of stored water due to turbidity (murkiness due to more sediment).

Flood risk

Flood risk (Figure 1.37) is a combination of:

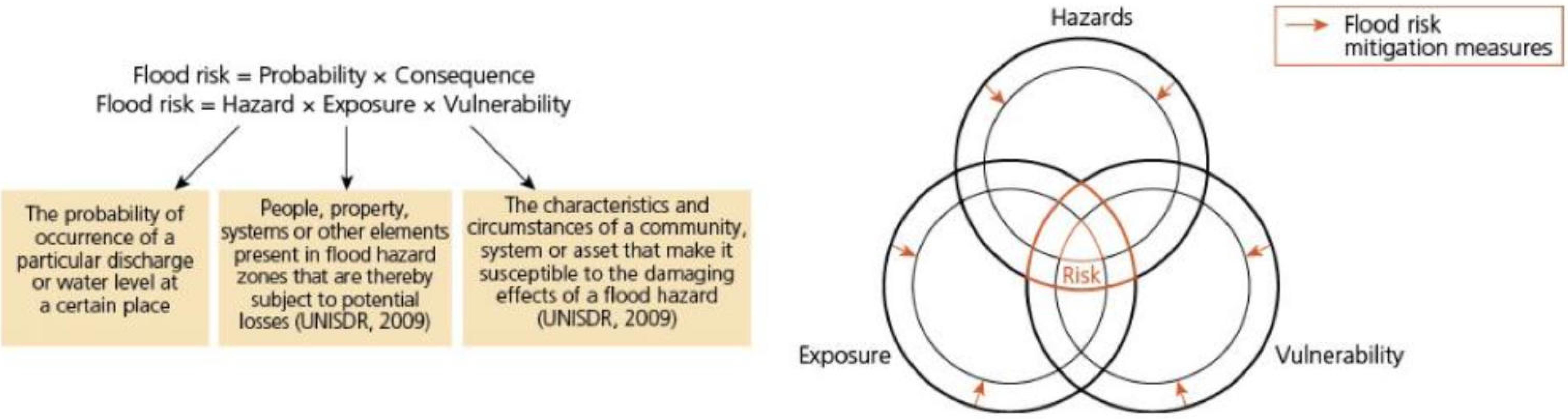
» hazard – the probability of occurrence
» exposure – a measure of the number of people or things that may be affected

» vulnerability – a measure of the potential for people and property to be affected.

Flood risk is dependent on there being:

» a source of flooding, in most cases a river
» a route, such as a river valley, for the flood water to take (a pathway)
» a receptor (people and property) that will be affected by the flood.

Therefore, the impact of a flood event will be considerably lower in a sparsely populated region (low exposure) and/or where people are able to evacuate quickly (low vulnerability), compared to in a densely populated region (high exposure) and/or where evacuation would be problematic (high vulnerability). This concept is known as the source-pathway-receptor model. In most countries there will be a system of **flood impact assessment**.



▲ **Figure 1.37** The concept of flood risk and its reduction, including Venn diagram

The strategies to manage river flooding: prediction and prevention

Prediction

In recent decades, flood forecasting and warning has become more accurate, due mainly to advances in weather satellites and the use of radar. This is particularly the case in developed countries. There is much less effective flood forecasting in most developing countries. However, there are interesting exceptions, such as Bangladesh. Most floods in Bangladesh originate in the Himalayas, giving authorities about 72 hours' warning. Disastrous floods, however, can often come as a surprise.

In the USA, National Weather Service forecasters rely on a network of almost 10,000 river gauges to monitor the discharge of rivers across the country. According to the **US Geological Survey**, flood prediction requires several types of data:

» the amount of rainfall occurring
» the type of storm producing the precipitation
» the rate of change in the discharge of the river/channel network
» the characteristics of the drainage basin.

The task then is to convey information as quickly as possible about the immediacy and severity of the flood risk to the people who are likely to be affected. For example, using the UK government website (https://gov.uk), allows you to check if you are:

» at immediate risk of flooding
» at risk of flooding in the next five days
» in an area that's likely to flood in the future.

Prevention

Traditionally, floods have been managed by methods of '**hard engineering**', such as dams, reservoirs, levées, straightened channels and flood-relief channels (Figure 1.38). Although hard engineering can be effective in reducing the risk in the locations selected, it may cause unexpected effects elsewhere in the drainage basin. Such negative consequences include increased sedimentation, bed and bank erosion, decreased water quality and loss of habitats. Levées are the most common form of river engineering. Over 4500 km of the Mississippi River has levées.

▲ **Figure 1.38** Hard engineering structures: (a) river defences on the River Thames, London; (b) levées in Zermatt, Switzerland

More recently, '**soft engineering**' measures have increasingly come to the fore. These techniques focus on working with natural processes and features rather than on attempting to control them. They include catchment management plans, river restoration and wetland conservation. **Flood abatement** involves reducing the amount of runoff in a drainage basin. This can be achieved through actions such as:

- reforestation, to slow down the movement of water in a drainage basin
- reseeding sparsely vegetated areas, to reduce evaporation losses and control soil erosion
- treatment of slopes by contour ploughing or terracing (Figure 1.39), which reduces surface flow
- clearance of sediment and other debris from streams, to increase river discharge
- preservation of natural water storage zones, such as lakes
- construction of small water- and sediment-holding areas
- comprehensive protection of vegetation from wildfires, **overgrazing** and clear-cutting of forests.

Flood diversion refers to the practice of allowing certain areas, such as wetlands and floodplains, to be flooded to a greater extent. Natural flooding may be increased through the use of flood-relief channels (diversion spillways) to direct more water into these areas during times of flood.

▲ **Figure 1.39** Terracing in mountainous terrain in Nepal – terracing increases agricultural production and reduces surface runoff

Land-use zoning

Hydrological responses to rainfall strongly depend on the local characteristics of the soil, such as water storage capacity and infiltration rates. The type and density of vegetation cover and the characteristics of the built environment are also very important in the hydrological response to rainfall.

Land-use zoning segregates land use on a floodplain into different areas by type of use. Most land-use zoning practices date from the mid-twentieth century, although earlier examples can be found. Carefully planned land-use zoning can reduce the number of premises and people at risk of flooding. For example, parks and gardens can be planned and designed to absorb and contain significant amounts of water during and after heavy rainfall. On a larger scale, **catchment management** of the whole catchment area to optimise the functioning of the catchment can use a source control approach that seeks to keep as much rainwater as possible retained where it falls, thus reducing flood peaks.

Floodplain land that floods regularly (once a year) could be used for pastoral agriculture, as animals can be moved to higher ground when there is a risk of flooding. Alternatively, such land could be used for recreational purposes, where there would be minimal damage to infrastructure. In many rapidly expanding urban areas, unplanned and poorly planned development has often outpaced the construction and improvement of drainage infrastructure.

Hazard-resistant building design

Hazard-resistant design has also become increasingly important. Hazard-resistant design (for example flood-proofing) includes any adjustments to buildings and their contents that help reduce losses. Table 1.4 shows adjustments to building and site design that should be considered when building in flood-risk areas. Incorporating flood-resistant construction measures during the construction and renovation of homes not only prevents damage, but also reduces post-flood stress.

▼ **Table 1.4** Four levels of adjustments to building and site design in flood-risk areas

Flood avoidance	Constructing a building and site in a way that minimises the chance of it flooding. For example: ● Building it higher than the flood level. ● Moving to an area that is not at risk of flooding.
Flood resistance	Constructing a building in a way that stops floodwater coming into the building and causing damage. For example: ● Using specialist flood doors and windows. ● Installing automatic anti-flood airbricks.
Flood resilience	Constructing a building in a way that reduces the impact of flooding should water get into the building, so that: ● permanent damage is avoided ● structural integrity is preserved ● the clear up and drying is made easier. For example:

	● Ensuring that electrics and sockets are positioned above likely flood levels. ● Installing tiles instead of fitted carpets.
Flood repairable	Constructing a building in a way that, should water get into the building, any damage caused by the flood can be repaired easily or replaced. (This is also a type of flood resilience.) For example: ● Using treated timber for flooring rather than untreated timber. ● Installing 'marine ply' kitchen units.

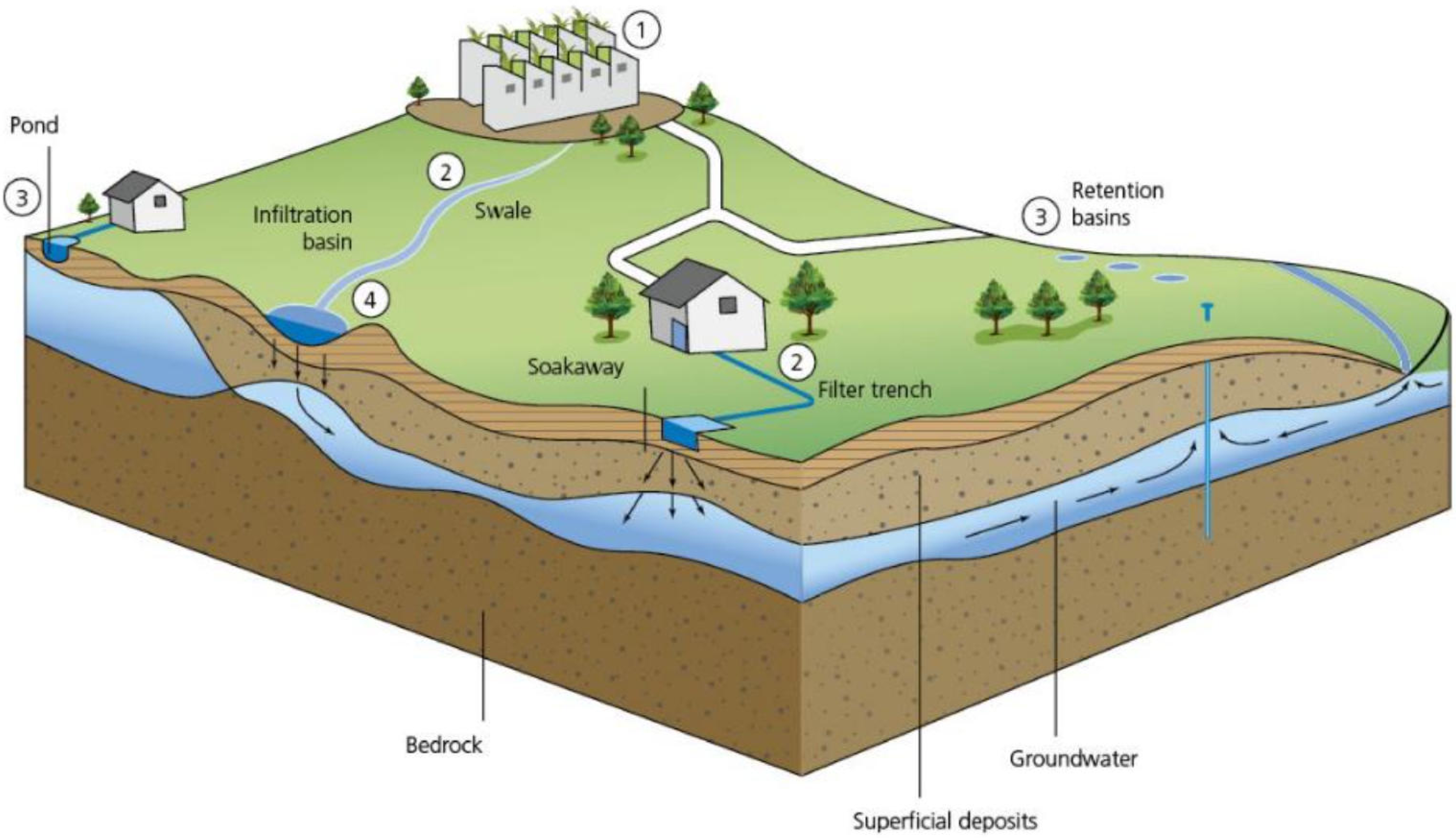
Sustainable drainage systems

Sustainable drainage systems (SuDS) are natural drainage methods for the drainage of surface water to nearby watercourses in urban areas. They provide an alternative to the direct channelling of surface water by pipes and sewers. They aim to:

» lower flow rates in watercourses
» increase surface water storage capacity
» improve water quality by reducing the transport of pollutants to water environments.

Figure 1.40 shows the different methods used in SuDS to decrease flow rates in watercourses and to improve water quality:

1 Source-control methods, such as intercepting runoff water on roofs for subsequent reuse or for storage and subsequent evapotranspiration (for example, green roofs), to reduce the volume of water entering watercourses.
2 Pre-treatment steps, such as vegetated swales (channels) or filter trenches that remove pollutants.
3 Retention systems that delay the discharge of surface water to watercourses by providing storage within ponds, retention basins or wetlands.
4 Infiltration systems, such as infiltration trenches and soakaways, that allow water to soak into the ground.



▲ **Figure 1.40** Sustainable drainage systems

Activities

1 Write a brief analysis of Figure 1.37 (page 21).
2 Which types of data does the US Geological Survey use for flood prediction?
3 Describe three methods of hard engineering.
4 a Define soft engineering.
b List four types of soft engineering.
5 Explain why appropriate land-use zoning can reduce flood risk.
6 What are sustainable drainage systems?

Detailed specific example

The 2019 Mississippi River flood

The Mississippi River drainage basin

The most prolonged and widespread flooding in US history occurred in 2019, with the Mississippi River drainage basin the most affected region in the USA. The Mississippi River basin drains one-third of the continental USA along with two Canadian provinces (Figure 1.41). The Upper Mississippi River, Arkansas River and Missouri River all drain into the Lower Mississippi River, and all recorded major flooding.

The flood events in the Mississippi basin caused estimated economic losses of $20 billion. The 2019 Mississippi River flood broke records set by previous flood events in 1973 and even 1927. The impacts of the floods in these two former years were colossal, and shape how the Mississippi River is managed now. Precipitation is increasing across this very wide area due to global warming, causing additional water to flow downriver towards the Gulf of Mexico. The shape of the river basin has been described as an inverted triangle, extending from the central Rocky Mountains to the west and central Appalachian Mountains to the east, to the Gulf of Mexico to the south. The main land use in the basin is cropland (58 per cent).

The 3371 km Mississippi River stretches from Minnesota to the Gulf of Mexico. It flows north to south through 18 degrees of latitude, with the climate type changing from cool temperate at its origin to sub-tropical at the mouth of the river. The river drains water and sediment from 31 US states, delivering both to the Gulf of Mexico.

▲ **Figure 1.41** A map of the Mississippi River basin

The causes and duration of the 2019 floods

A lethal combination of heavy spring rain, rising temperatures and melting snow overwhelmed waterways in the drainage basin. The 12 months from May 2018 to April 2019 were the wettest year-long period in records dating back to 1895. In the parts of the basin subject to heavy annual snow accumulation, snow cover has generally decreased. This is due to:

- warmer temperatures causing earlier melting
- more precipitation falling as rain rather than snow.

There is also increasing evidence to suggest that the persistent **jet stream** behaviour behind these record-setting precipitation events is most likely due to human-induced climate change. Red River Landing is located on the Mississippi River near the confluence of the Mississippi and Red Rivers, and Figure 1.42 shows the number of days above flood stage at Red River Landing for the floods of 1927, 1973, 2011 and 2019.

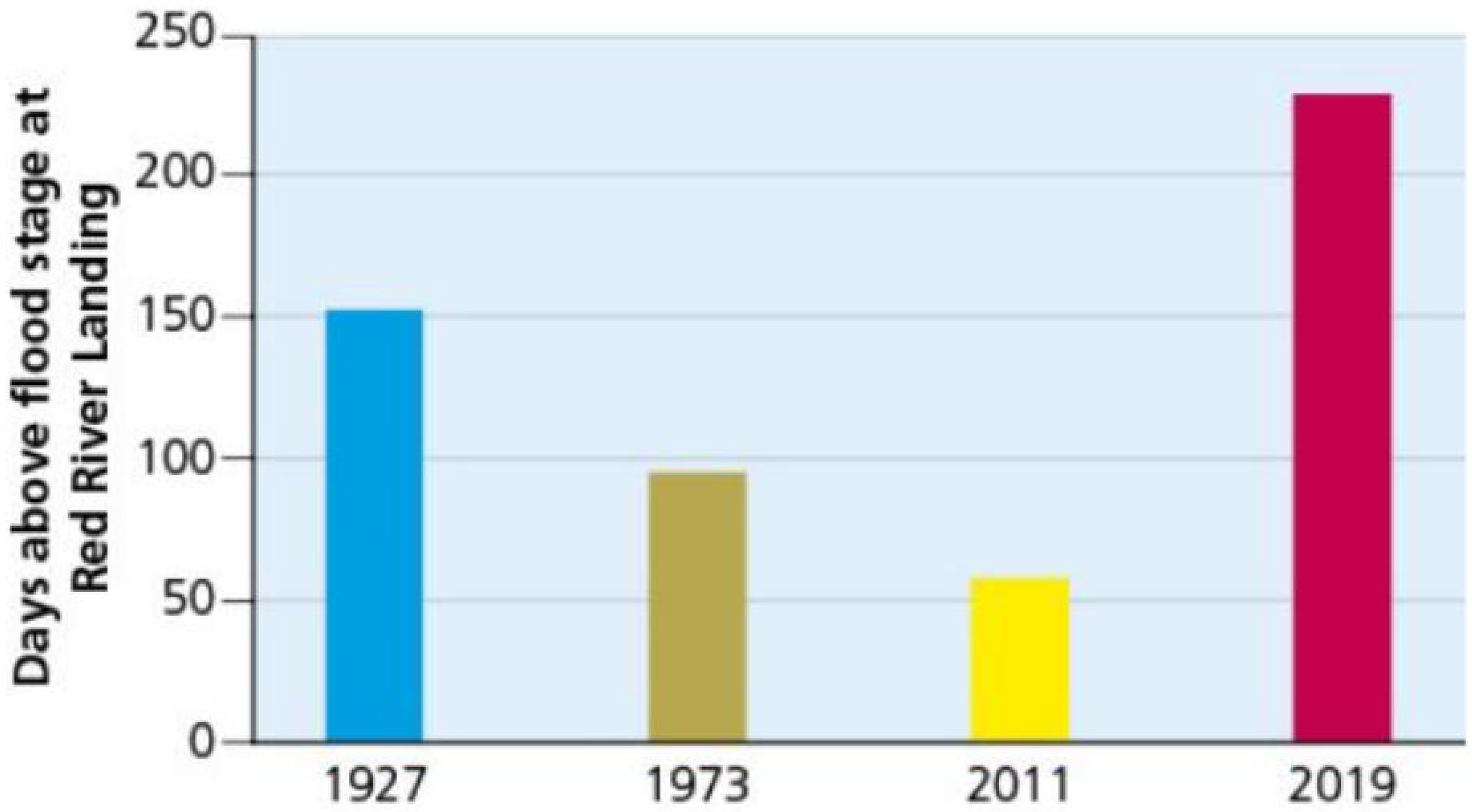
There were three major periods of flooding between March and September 2019. Many areas in the basin were flooded more than once.

- In March, part of the US Midwest and the northern states were impacted by a 'bomb cyclone' that precipitated huge amounts of snow and rain in a short time period.
- Peak flooding from spring to July was the most devastating event.
- A final period of flooding occurred in late summer/early autumn.

However, the US Army Corps of Engineers had been working since early November 2018 to manage flood waters and prevent flooding in communities across the Mississippi River Valley. While communities upriver had experienced devastating flooding, those in and around Louisiana had been largely protected by the Mississippi River and tributaries system.

Between January and June 2019, precipitation was as high as 150–250 per cent above normal in much of the drainage basin. For much of the region it was the wettest period in 124 years. By early July 2019:

- the Mississippi River had been above flood stage for 235 days, the longest period in recorded history



▲ **Figure 1.42** The number of days above flood stage at Red River Landing for the floods of 1927, 1973, 2011 and 2019

- nearly 210 trillion gallons of water had flowed down the Mississippi River since the beginning of the year – 64 per cent above the 10-year average.

For the first time in its almost 100-year history, the Bonnet Carre Spillway was opened twice in back-to-back years to relieve pressure on levées and prevent devastating flooding over a large area. In 2019, the Bonnet Carre Spillway was open for 123 days.

The effects of the flooding

- The floods caused nearly 20 million acres of farmland to go unplanted.
- Around 14 million people were displaced for some period of time.
- More than 550,000 acres in the Mississippi delta, almost half of it farmland, was impacted by flooding and stagnant water.
- St Louis Harbour was closed for 38 consecutive days between May and June 2019. From March to June, an estimated 6.3 million tonnes of grains worth almost $1 billion went unshipped due to disruptions to barge traffic.
- The flooding led river authorities to believe many river passages unsafe due to strong currents and obstructions along the river bed. They closed multiple locks, suspended traffic on certain sections, set tow size restrictions and in some areas limited passage under bridges to daylight hours.
- In Louisiana, the seafood industry suffered severe disruption. The influx of a much higher than normal level of fresh water impacted the Gulf of Mexico ecosystem and decimated species such as shrimp and oyster.

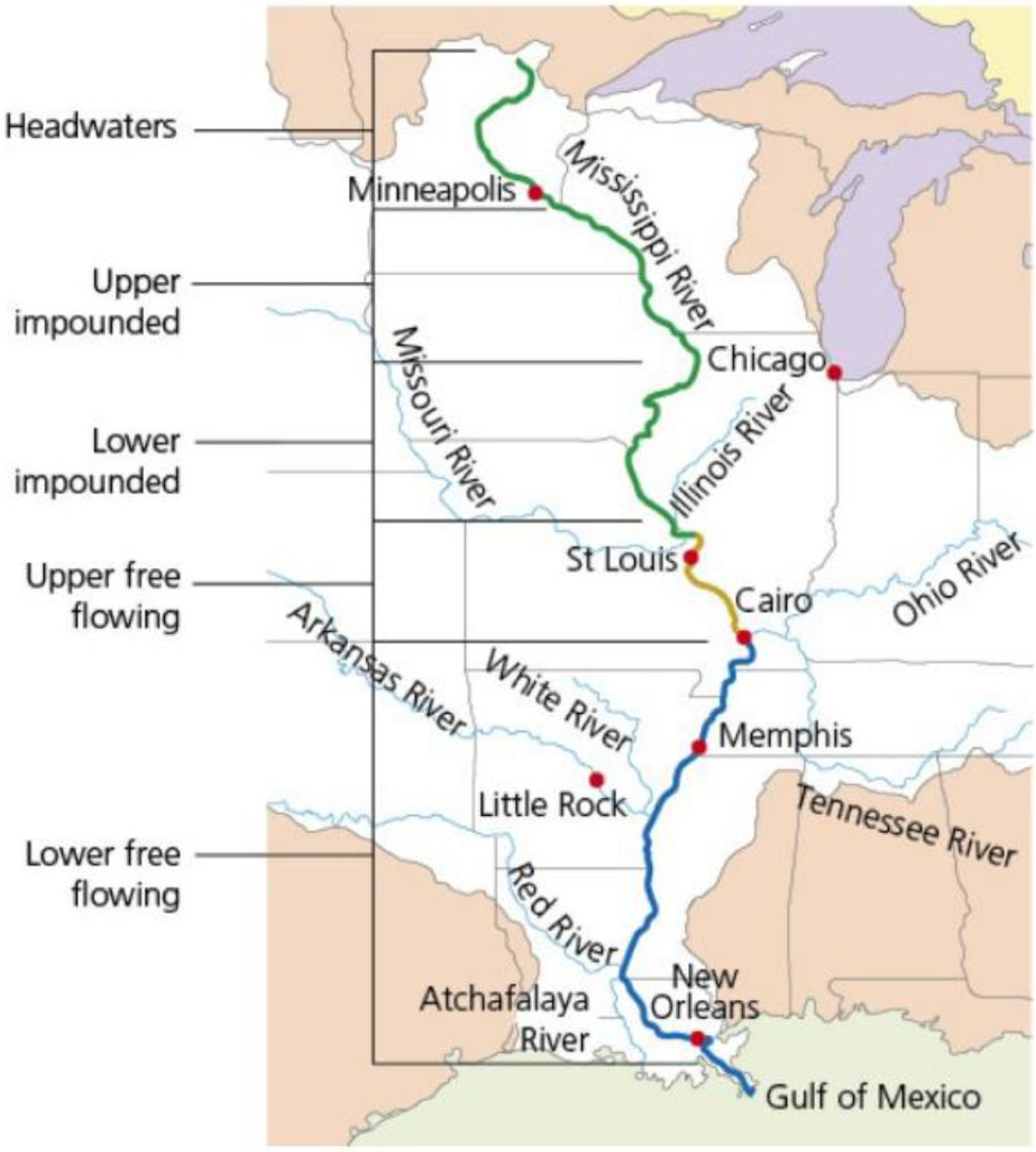
Flood protection measures

Although flood protection measures have steadily improved through a number of phases along the Mississippi, and in more recent times have involved soft engineering as well as hard engineering, the USA was taken by surprise by both the scale and duration of the 2019 floods. Hydrologists argue that lessons must be learned, stressing the urgent need to consider managing the Mississippi River more holistically to deal with the more frequent and intense floods expected in the future. Communities across coastal Louisiana also fear water coming up from the Gulf of Mexico.

Figure 1.43 shows the reaches of the Mississippi River and the alterations made to the natural river over time. The headwater reach extends from the river's source at Lake Itasca to Coon Rapids, Minnesota. Through much of this reach, the river flows through wetlands and forest. Alterations in the impounded reaches focused on making the river navigable for commercial shipping and on flood control to allow development of the floodplain for agriculture and urban areas. The process of impoundment began in the 1930s. Levée construction disconnected a substantial portion of the historic floodplain from the river.

The magnitude and range of alteration is greatest in the free-flowing reaches of the Mississippi River. This section of the river begins just upstream of its confluence with the Missouri River. Dikes (also called wing dams or wing dikes) are used to direct the flow of water to maintain the alignment and depth of the navigation channel. Dikes are usually constructed in groups of three or more (dike fields), with each successive downstream dike extending further into the channel. Revetted banks, which are reinforced with concrete or other supportive material, protect the outside of meanders from erosion. Levées dominate the river banks and have been incrementally raised over time in response to large flood risks. As in the impounded reach, the levées disconnect the river from much of the historic floodplain. In urban areas along the river the levées are often replaced by floodwalls, which use less land. Dredging occurs when required and is used in various stretches of the river's course all of the time.

There are of course many other examples of dams and other alterations on the tributaries that feed into the Mississippi River. These include on the Missouri, Ohio and Tennessee Rivers.



▲ **Figure 1.43** The reaches of the Mississippi River

Sustainable management strategies

Restoring floodplains is an important strategy for communities attempting to manage annual flooding along the Mississippi River. As the river proceeds downstream from its headwaters, an increasing proportion of the floodplain is disconnected from the river. Disconnected floodplain refers to the portion of the historic floodplain that is now separated from the river by levées.

Flood preventative land-use zoning is widely recognised as an important soft engineering technique. The main problem is that, without huge expense and disruption, this process can only function at a slow speed. Areas of land on floodplains have been purchased by federal and local governments and by NGOs to prevent further development and to implement nature-based solutions for flood control. Protecting existing wetlands through zoning is often an important first step. Differences of opinion between the federal government and local governments over land use in floodplains have been a regular occurrence over the years.

Pairing levées with nature-based solutions is viewed as the best way forward by many hydrologists working along the Mississippi River. Nature-based solutions mean that levées are better prepared to resist floods. The concept of 'levée setbacks' is gaining increasing attention. This involves relocating levées further away from a river and creating space for nature-based solutions between the river and the relocated levée.

The Nature Conservancy has played a significant role in promoting nature-based solutions:

- In the Atchafalaya River basin in Louisiana, a million acres of wetland provide a critical natural habitat. The Nature Conservancy is working to ensure the natural benefits of flooding by protecting the existing floodplain forest, planting new trees and restoring natural hydrology where possible. It has shaved the tops off sections of canal levées and cut notches in river banks to allow floodwaters to once again flow through the wetlands.
- In Iowa, The Nature Conservancy is working with farmers to promote in-field practices such as prairie strips. These are sections of native prairie vegetation planted within crop fields to slow runoff. The planting of cover crops has also been encouraged. Cover crops are planted during the off-season to hold soil in place and break up ground compacted by farm machinery. Additional practices, including no-till farming or only partially tilling fields, create a more absorbent soil surface that holds in moisture during the drier months and reduces runoff when conditions are wet.
- In Missouri, flooding periodically affects almost every county. Communities are advised on the benefits of nature-based solutions to address flood risk. These include planting trees to replace areas of turf grass on floodplains (as the latter has a very low level of water percolation when the land is sloping), and building bio-retention ponds to hold some of the runoff from flooding.

Activities

1 Describe the area covered by the Mississippi River basin.
2 Discuss the causes and consequences of the 2019 floods.
3 Explain two traditional methods of flood protection used along the Mississippi River.
4 Outline two nature-based strategies that have been used in the Mississippi River basin.

River pollution

The human causes of river pollution

Pollution is sometimes referred to as the 'invisible water crisis'. The composition of surface and underground water has an impact on **water quality**, and this depends on two broad factors:

» Natural conditions in the drainage basin
» Human intervention.

Human intervention is mainly in the form of **pollution**. Water pollution comes from a number of sources (Figure 1.44), including:

» contamination by agricultural runoff, particularly from factory farming, where livestock are kept on impermeable surfaces as opposed to grazing in fields
» industrial pollution of rivers and other water bodies
» urban runoff carrying pollutants from cars, factories and other sources
» untreated sewage, which has increased in many areas as population has grown.

▲ **Figure 1.44** A polluted river in Christchurch, New Zealand

Countries in all income groups are showing signs of risks related to water quality. Low levels of wastewater treatment is the main cause of poor water quality in low-income countries (LICs), while in high-income countries (HICs) runoff from agriculture is a more serious problem. A serious constraint in the efforts to improve water quality is that detailed water quality data remains sparse, largely due to weak monitoring and reporting capacity.

In 2019, the World Bank identified three types of pollution:

» Pollutants of poverty: Developing countries have lacked the finance to provide adequate water treatment facilities.
» Pollutants of growing prosperity: The use of nitrogen as a fertiliser has risen more than 700 per cent since 1960. Much of this growth has occurred in Asia.
» Emerging pollutants: Microplastic pollution has grown rapidly throughout the world's fresh water sources. Emerging pollutants are largely associated with developed economies.

Water pollution is forecast to intensify over the next 20 years or so and to present a serious threat to sustainable development. The list of contaminants of concern is increasing and exposure to pollutants is likely to increase significantly in low- and lower-middle-income countries. In these countries, pollution will be driven by population growth, economic growth and the lack of wastewater treatment. The increase in pollution is projected to be particularly strong in some African countries.

The 2020 UN World Water Development Report found that opportunities were being missed to use water projects to cut **greenhouse gas** (GHG) emissions while improving access to clean water. Wastewater is responsible for about 5 per cent of all GHG emissions globally. Processing sewage can turn wastewater from a source of GHGs to a source of clean energy if the methane is captured and used in place of natural gas. Currently 80–90 per cent of wastewater is discharged to the environment without treatment.

The impacts of river pollution

Water pollution reduces the ability of a water source to provide the quality of water that it would otherwise provide. Each year, more than 80 per cent of the world's wastewater is released to the environment without being collected or treated. This pollutes the environment and wastes a renewable resource.

While rivers in more affluent countries have steadily become cleaner in recent decades, the reverse has been true in much of the developing world. A significant reason for this is the impact of globalisation, in terms of outsourcing production to developing countries. Rivers in Asia are the most polluted. For example, each day around 800 million litres of sewage are drained into the Yamuna River that flows through Delhi. For many people, the only alternative to using this water for drinking and cooking is to turn to water vendors, who sell tap water at greatly inflated prices. According to a 2018 survey by the Delhi government, 44 per cent of residents in Delhi's lowest-income areas rely on bottled water.

Sewage and runoff from farms frequently contain nutrients such as nitrogen and phosphorus. These nutrients cause excessive aquatic plant growth (**eutrophication**), which can have a range of adverse ecological effects.

Poor water quality can have a very direct and massive impact on people's lives. Most seriously it can result in waterborne diseases, which can have significant health implications. Apart from health, the economic wellbeing of individuals and families is also likely to be adversely affected. If large numbers of people are affected by waterborne diseases, this will impact on national productivity. Diseases related to poor quality water include:

- bilharzia, where snails transmit flatworms to people, causing internal organ damage
- malaria and yellow fever, both caused by mosquitoes which breed around water
- cholera, which causes extreme diarrhoea.

Although most people in developed countries think their water supplies are clean and healthy, there is growing concern about traces of potentially dangerous medicines that may be contaminating tap water. Easily dissolved in water, these remain highly toxic when leaving the body and are hard to destroy in water treatment plants. Traces of medicines in water come from manufacturing plants, the livestock industry and domestic households. Many of the more than 4000 prescription medicines used for animal and human health ultimately find their way into the environment, including Bisphenol-A (BRA), antibiotics and opiates.

Activities

1 **a** Define pollution.
  b List three sources of pollution.
2 Comment on the types of pollution (related to income) identified by the World Bank.
3 Name three diseases related to poor water quality.

The strategies and techniques used to manage river pollution

River pollution management strategies can be grouped into three categories:

- Reducing human activities that produce pollution that ends up in rivers.
- Reducing the release of pollution into rivers.
- Removing pollution from rivers and restoring ecosystems.

Reducing human activities that produce pollution

Pollutants draining into rivers from agricultural activities are a major problem in many drainage basins. Agriculture is the single largest cause of river pollution in the UK and many other countries. The more intensive the agricultural practices are, such as intensive poultry units (IPUs), the higher the level of pollutants washed into rivers. Farming using organic principles ('green' farming) substantially reduces water pollution, using methods including the following:

- No-till and low-till farming: By disturbing the soil less, there is less runoff from rainwater. This reduces both sedimentation and pollution from runoff.
- Reducing the use of chemical inputs: Sustainable farming aims to reduce as far as possible the use of artificial fertilisers, insecticides and weed killers. Fertiliser pollution increases the level of nitrates and phosphates in river water, resulting in the growth of algae that form a bloom over the water surface.
- Preventing animal waste from leaching into groundwater and causing contamination.
- Agroforestry: Combining farmland with the planting of trees, shrubs and hedges. It can also involve the use of riparian buffer strips – vegetation planted alongside a river to act as a buffer for runoff.
- Less intensive animal rearing: Meaning that animals will not require the routine use of antibiotics.

Point-source pollution from industrial premises located beside rivers can be a heavy source of water pollution. This depends very much on government regulations and monitoring. In general, the combination of regulation and monitoring is more extensive in HICs, which have greater resources to establish effective monitoring systems. At the domestic/household level, limiting the use of detergents containing phosphates is an effective way to reduce a household's pollution footprint.

An increasing awareness of how salt harms fresh water sources has seen some local authorities in the USA and elsewhere significantly cut back on the use of road salt to counter potential freezing conditions on road surfaces. Such local authorities are now using 'salting' more carefully and selectively. Road salt is generally the dominant source of chloride in regions that experience a significant level of freezing in winter. Unlike other pollutants, chloride doesn't break down in water over time, and levels of chloride have increased by more than a third since the late 1980s across the entire Upper Mississippi River basin.

Other reasons for the increased level of chloride in river water include:

» salt from water softeners
» potassium chloride fertiliser.

Reducing the release of pollution into rivers

Wastewater treatment is a vital process that helps to ensure water supply systems remain safe and sustainable. Pollution from unmanaged wastewater remains a pressing global challenge. Wastewater treatment involves removing pollutants from wastewater through physical, chemical or biological processes. The more efficient these processes are, the cleaner the river water becomes. The infrastructure required for large-scale effective wastewater treatment is expensive. Therefore it is not surprising that there is a high correlation between the GDP per capita of a country and the quality and extent of its wastewater systems.

However, cost is not the only obstacle. Wastewater treatment processes contribute as much to global GHG emissions as the global aviation industry. Conserving water by reducing per capita water use in a practical way, without compromising personal health, would relieve pressure on existing systems and reduce GHG emissions.

When stormwater is not properly managed, it can pick up pollutants from the surfaces it flows over. Effective stormwater management therefore reduces the amount of pollutants contaminating rivers, by methods including:

» permeable pavements to allow stormwater to infiltrate through porous surfaces into the soil and groundwater
» roadside curb 'cuts' to allow road runoff to be directed into pervious areas
» vegetated filter strips – bands of dense vegetation through which runoff is directed
» constructed wetlands designed to operate in a similar way to natural wetlands.

Altering human activities to reduce the volume and range of pollutants entering rivers is normally achieved by a combination of government legislation and education campaigns. Raising both public and commercial awareness of the causes and effects of river pollution can result in significant benefits.

Removing pollution from rivers

A range of remedial actions can be employed to revive a polluted river:

» The clearing and stabilisation of river banks to minimise further direct pollution from an area.
» Physical pollution, much of it in the form of litter, can end up in rivers from a variety of sources. Volunteer organisations in a wide variety of locations around the world carry out regular litter picks. A significant proportion of such river litter is in the form of plastic pollution (Figure 1.45). A high proportion of plastic reaching the sea comes from rivers.
» Physically removing algae blooms, which have been created by excessive nutrients (for example fertiliser runoff) being washed into a river, may be an essential activity in the process of returning a river to a healthy state.
» Pumping air into the water where the environmental state of a river is extremely poor, using floating aerators. This can prove effective in combination with other clean-up activities. Oxygen is essential for the balance and function of fresh water ecosystems. Aeration methods can be top down (surface aeration) or bottom up (benthic aeration). Some water bodies require constant aeration, while others require only emergency aeration during high-risk periods.
» Mechanical dredging to remove contaminated sediments, which will also have the benefit of increasing river flow.

▲ **Figure 1.45** Plastic pollution along the banks of the lower Danube River, Romania

Activities

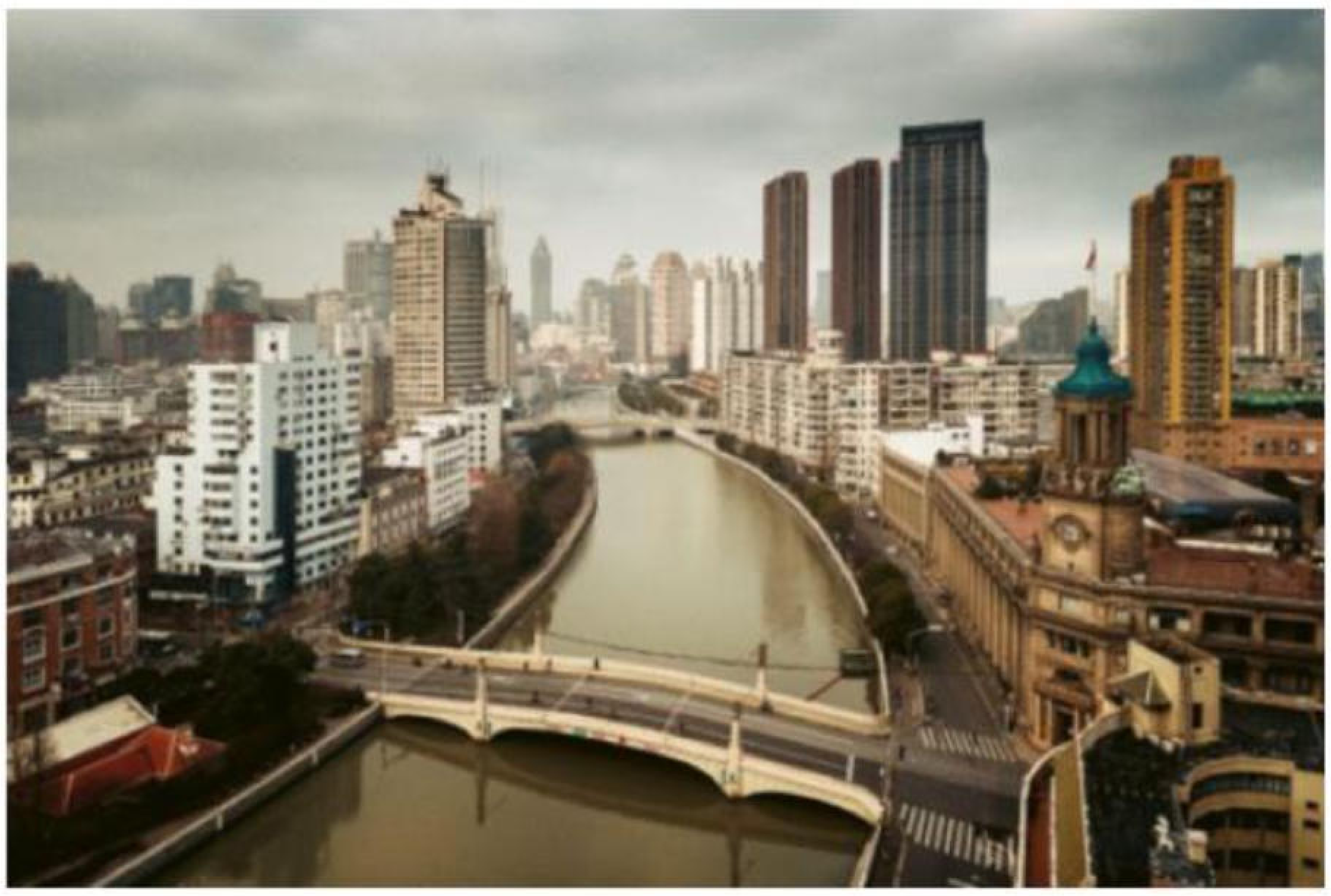
1 Describe two changes that could be made to agricultural practices to reduce polluted runoff into rivers.
2 How can good stormwater management reduce river pollution?

Detailed specific example

Restoring Shanghai's Suzhou Creek

Location, causes and consequences

Suzhou Creek (Figure 1.46) is a part of the Yangtze River drainage basin. Its source is Taihu Lake, from where it winds 125 km, of which almost 53 km flows through Shanghai, before its confluence with the larger Huangpu River. Shanghai is one of the world's largest cities. It is located at the mouth of the Yangtze River and bounded to the east by the East China Sea.



▲ **Figure 1.46** Suzhou Creek flowing through central Shanghai

Beginning in the early 1900s, the waters of Suzhou Creek deteriorated, mainly due to increasing population and the expansion of industrial activities. Domestic sewage and industrial wastewater were discharged directly into the river, gradually polluting the water quality. By the late 1970s the entire river was heavily polluted. Suzhou's fish and shrimp populations became extinct in the 1980s. By this time it had become the most polluted water body in Shanghai. There was a high level of visual pollution and frequent complaints about the foul odour released by the mix of pollutants. Algal blooms in early summer had become a common occurrence.

▼ **Table 1.5** The increasing population of Shanghai

Year	Population
2024	29.9 million
2010	20.3 million
1990	8.6 million
1970	6.1 million

An important transport route for centuries, Suzhou Creek had become overloaded with boats transporting a wide array of products. Apart from its transport function, the waterway also provided vital flood control and drainage functions for the area, and water for farmland irrigation and nearby factories. Over a long period, the river had deteriorated due to:

- raw industrial pollution
- urban wastewater
- spills and waste from boats
- human waste.

The situation had become so bad that the river was nicknamed the 'black and stink'. At this point, Suzhou Creek:

- failed to meet China's lowest national water quality standard (Class V)
- had become a public health hazard due to the risks of diseases such as cholera, typhoid and dysentery.

Low-income communities were most vulnerable to the hazards posed by the river as they lived, and their children played, near the creek.

Restoration in three phases

In an attempt to tackle this major water pollution problem, in 1996 the Shanghai Municipal Government and the Asian Development Bank (ADB) entered a long-term partnership to restore the creek. The Economic and Social Development Plan for Shanghai was adopted to begin the 12-year Suzhou Creek Rehabilitation Project. The first phase of this project was launched in 1998 and completed in 2003. It involved:

- the reduction of sewage discharge into the river by developing systems to divert sewage away from the creek

- the construction of a pumping station to flush the creek and input oxygen into the dead water body
- the installation of a water lock between the Huangpu River and Suzhou Creek
- the construction of a wastewater treatment plant, with a capacity of 400,000 m^3 a day
- the installation of solid waste collection wharves.

The second phase followed on immediately from the first phase and was completed in 2005. Its objectives were:

- to maintain and improve the water quality (building on the improvements in the first phase)
- to extend the cleaning to its six tributaries
- to construct embankments to develop large areas of green space along the river banks.

The third phase (2006–08) continued the emphasis on improving riverside accessibility for surrounding residents. Public use of riverfront space, appropriate distribution of a riverfront greenbelt and regulation of riverfront buildings were included in the Suzhou Creek Landscape Planning. This final phase emphasised the social and environmental benefits of the project to local residents.

The combination of these measures removed wastewater from the creek and kept it out. Overall, the project benefited about 3 million people living in proximity to the creek. It improved the environment of a considerable area of Shanghai by:

- improving sanitation services
- reducing serious risks to public health
- providing greater access to parks and green spaces along the river banks.

Public surveys conducted after completion of the project showed that satisfaction with the city's environment had increased from 12 per cent in 2000 to 71 per cent in 2003. Satisfaction with water quality had risen from 12 per cent in 2000 to 76 per cent in 2003.

Finance and project partners

The initiation and ultimate success of the project depended on collaboration between a number of partner organisations. The total cost of the project was US$876 million, comprised of:

- a US$300 million loan from the Asian Development Bank
- US$325.4 million from the State Development Bank
- US$132.5 million from the Ministry of Finance
- US$62.7 million from the Shanghai Municipal Government
- US$55.4 million from district and county governments.

Sustainable management

The Chinese government has stressed environmental protection and sustainable management as a national strategy under its 'Ecological Civilisation' programme. UN HABITAT has also recognised the regeneration of Suzhou Creek as a success.

Major sustainability measures include:

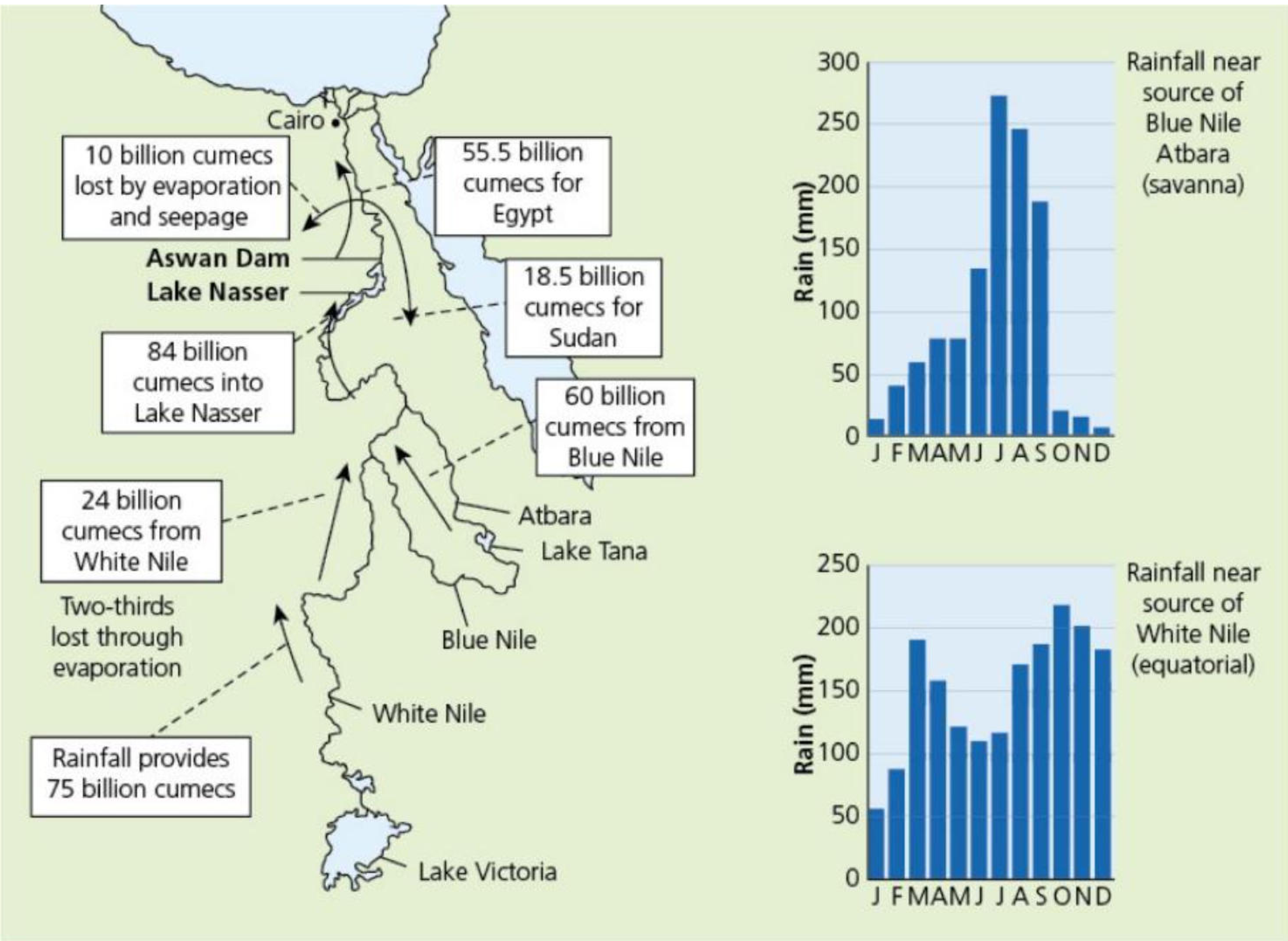
- Education and legislation to reduce pollutants entering the river system.
- A 'River Chief' system of responsibility at all levels of government has been established for the management and protection of rivers.
- Maintaining a sufficient level of investment to monitor water quality and maintain the necessary quality of infrastructure. Continuous monitoring of pollution levels in the drainage basin as a whole is vitally important for successful sustainability.

Activities

1. Why is Suzhou Creek regarded as an important waterway?
2. How did the river become so polluted?
3. Describe the three phases in the restoration of the river.
4. Suggest why so many organisations were involved in the restoration project.

Practice questions

1. Figure 1.47 shows the inflows and outflows on the Nile water and rainfall graphs for the source areas.

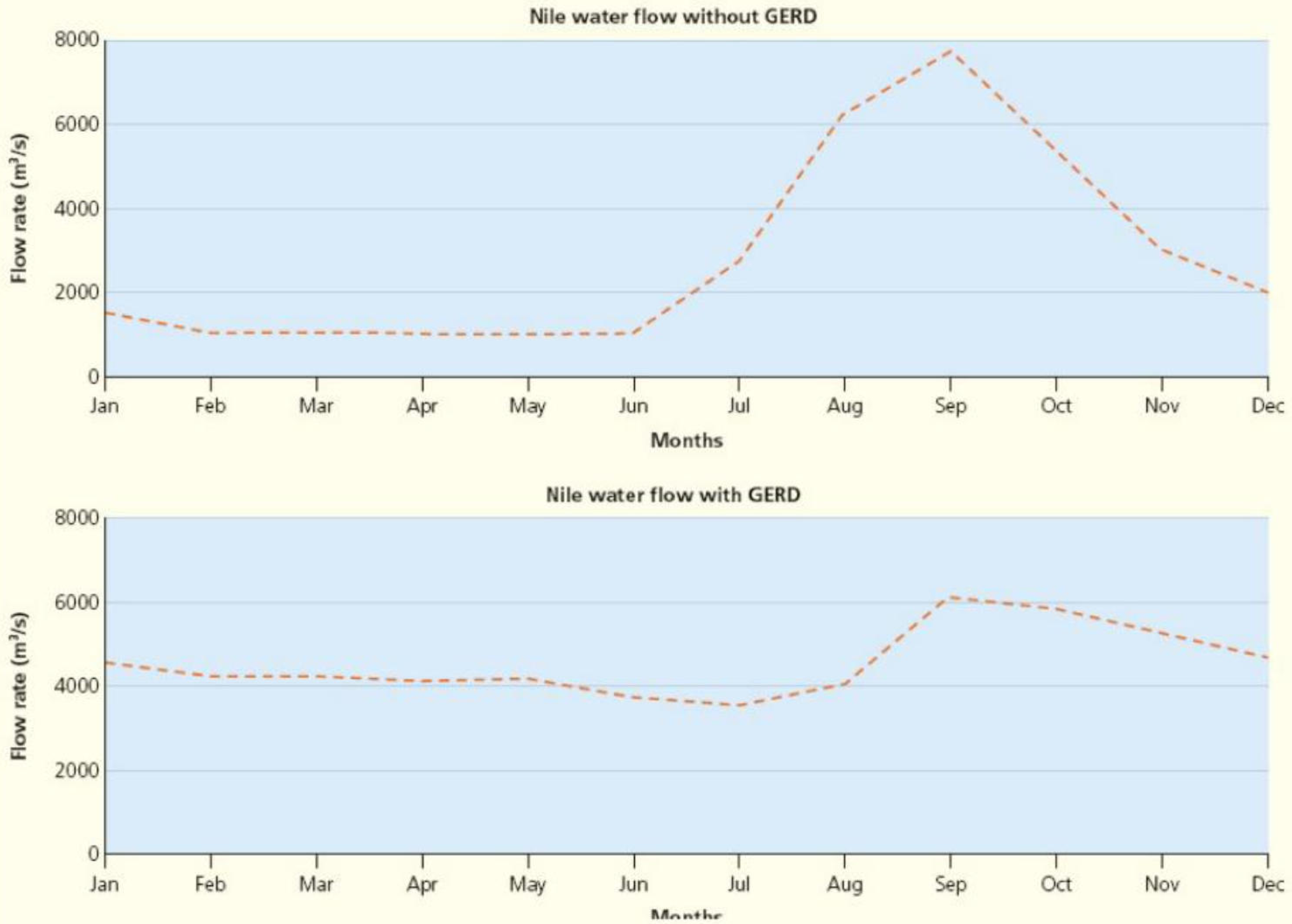


▲ **Figure 1.47** The inflows and outflows on the Nile water and rainfall graphs for the source areas

- **a** State the amount of water that:
  - **i** initially comes from the White Nile [1]
  - **ii** joins the Blue Nile. [1]
- **b** Describe the rainfall pattern for the source of the Blue Nile. [3]

2 Figure 1.48 shows the changes in the flow of water in the Nile before and after the building of the Grand Ethiopian Renaissance Dam (GERD), completed in 2020.

- **a** State the maximum flow in the Nile:
  - **i** without the GERD [1]
  - **ii** with the GERD. [1]
- **b** Compare the flow of the Blue Nile without the GERD and with the GERD. [3]



▲ **Figure 1.48** The changes in the flow of water in the Nile before and after the building of the GERD, completed in 2020